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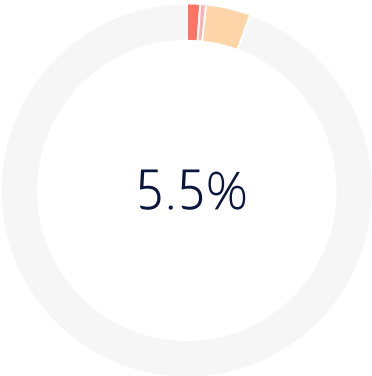

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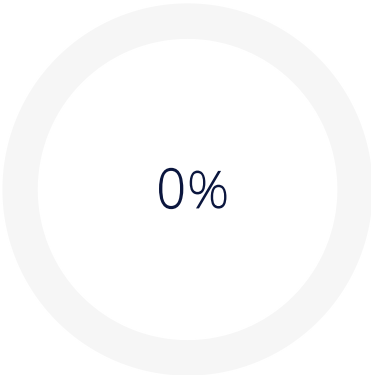

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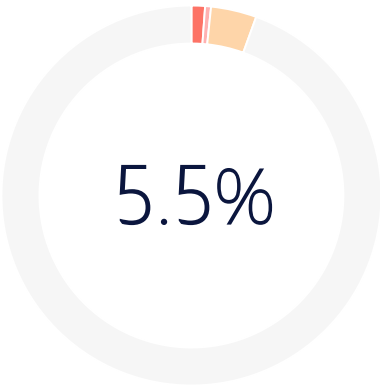

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A Comparative Study of Swin Transformer and ResNet50 for Automated Detection of Age-Related Macular Degeneration Using Fundus Images

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Abstract—Age-related macular degeneration (AMD) is a major reason for irreversible loss of vision in the central portion of the visual field and is a significant problem in people over the age of 75 years worldwide. The problem of early diagnosis is still a big issue due to the interpretation of retinal images and the lack of availability of expert ophthalmologists. To handle the problem of data imbalance and improve the generalization of the model, the Synthetic Minority Oversampling Technique (SMOTE) is used in this study to generate synthetic images of the minority class, i.e., age-related macular degeneration. The study focuses on the application of deep learning models, specifically the Swin Transformer and ResNet50, for the detection of age-related macular degeneration from retinal images. The Swin Transformer is based on the hierarchical window-based self-attention mechanism, which is effective in extracting global features from the images, and ResNet50 is a deep convolutional neural network that is effective in extracting local spatial features and is computationally efficient even on small datasets.

Index Terms—Age-related macular degeneration, retinal fundus imaging, deep learning, convolutional neural network, Swin Transformer, ResNet50, SMOTE.

I. INTRODUCTION

Age-related Macular Degeneration (AMD) is a progressive retinal disorder and one of the leading causes of irreversible central vision loss among older adults worldwide [1], [2]. Recent global estimates suggest that nearly 196 million people were living with AMD in 2020, and prevalence is expected to climb substantially as populations continue to age [2]. The condition primarily affects the macula, disrupting high-resolution visual functions such as reading, recognizing faces, and driving, which creates considerable socioeconomic and healthcare burdens [1]. Although early diagnosis can help in slowing down the disease's progression, reviewing the fundus images manually is a laborious task and an inherently subjective one since it relies on clinical experience, leading to variable results [3].

However, with advancements in Artificial Intelligence, it has become possible to develop automated diagnostic systems to assist ophthalmologists in their tasks. Convolutional Neural Networks (CNNs), particularly deep architectures like ResNet50, have shown strong results in retinal image analysis and AMD classification by learning layered feature representations directly from fundus images [4], [7]. The residual learning framework at the core of ResNet50 allows deep networks to train efficiently and has proven reliable even when working with relatively small medical datasets [7], [8]. That said, CNN-based models have a well-known limitation: they struggle to capture long-range dependencies and broader contextual relationships, both of which matter when identifying the subtle and spatially distributed features associated with AMD.

Vision Transformers (ViTs) were developed to address exactly this gap. By using self-attention mechanisms, ViTs can model global context across an entire image, and recent studies have shown they can match or even outperform CNNs on a range of medical imaging tasks [4]. The Swin Transformer builds on this foundation with a hierarchical design and shifted window-based attention, which enables efficient multi-scale feature extraction without becoming computationally prohibitive [5]. Even so, there has been relatively little research comparing CNN-based models like ResNet50 directly against transformer-based models like the Swin Transformer in the specific context of AMD detection.

This study aims to fill that gap by evaluating and comparing the performance of ResNet50 and the Swin Transformer in classifying AMD from retinal fundus images. Both models are trained on a balanced dataset prepared using SMOTE-based oversampling to handle class imbalance. The evaluation draws on accuracy, precision, recall, F1-score, and AUC metrics to offer a clear picture of how these two approaches compare in terms of model complexity, generalization, and overall diagnostic performance.

II. RELATED WORK

The conventional method of identifying AMDs has always involved manual analysis of fundus photos, OCT imaging, and angiography. Even as these approaches have been effective, they do present some problems,

including subjectivity of interpretation, inefficiency, and the need for skilled personnel. It is, therefore, natural that scientists have looked to automation techniques for assistance.

The initial impetus for automating AMD detection was driven by CNN models. A notable contribution was made by DeepSeeNet, a deep learning model trained on the large-scale AREDS fundus dataset, which demonstrated that the technology could be effective enough to equal specialist AMD grading accuracy [9]. Similarly, Burlina et al., performed one of the first AMD classification studies utilizing CNN algorithms, based on AREDS image data, achieving impressive results which provided validation of the approach to automated screening [10], [11]. Some other CNN models focused on the neovascular (wet) type of AMD specifically; Keel et al., successfully tested the sensitivity and specificity of their deep learning approach using clinical OCT and fundus data for diagnosis [12]. In continuation of this progress, more advanced CNN models incorporated scale-adaptive techniques, hybrid feature extractors, and improved pre-processing pipelines to enhance performance, as evidenced by the research of El-Den et al. [13] and the AMDNet23 architecture introduced by Ali et al. [14]. Through most of this process, ResNet50 became a preferred choice because of its residual learning structure, facilitating deep learning while ensuring stability [7], [8].

Data deficiency and class imbalance have been constant issue in studies about AMD, and some research teams attempted to solve this problem by means of creating synthetic data and oversampling. Oliveira et al. used StyleGAN2 to create synthetic fundus images, finding out that such an approach was able to enhance the robustness of models trained on AMD and other retinal diseases [15].

When transformers started to make waves in the medical imaging domain, Vision Transformer (ViT) gained interest among researchers. Akerman et al. evaluated transformer-based models across a range of ophthalmic tasks and found that ViTs can surpass CNNs when it comes to capturing global retinal structure and long-range pathology patterns, including those tied to AMD [16]. Despite this growing body of work, direct comparisons between ResNet50 and modern hierarchical transformer architectures for AMD detection remain scarce, and it is precisely this gap that the present study sets out to address.

III. DATASET AND PREPROCESSING

A. Dataset Description

The ADAM dataset, which comprises 400 color fundus images, will be used in this study. The images in this dataset belong to one of two classes: AMD and Non-AMD. The total number of images in this dataset is imbalanced, with 89 images belonging to the AMD class and 311 belonging to the Non-AMD class. The images were arranged in two separate folders according to their respective classes.

In this study, to increase diversity in the samples, augmentation was applied to the existing images, and then the augmented images were stored physically in directory structures based on training, validation, and testing sets. After augmentation, each set contained 1335 AMD images and 3110 Non-AMD images, with equal distribution in all three sets.

The resolution of all the images was adjusted to a standard size of 224×224 pixels, as required by ResNet50 and Swin Transformer architectures for input images. The pixel intensity was then standardized based on standard statistics for the ImageNet database, which is applicable to all pre-trained models.

B. Data Cleaning and Preprocessing

Prior to augmentation, the following preprocessing steps were performed on the original ADAM images:

- Image Resizing: All the fundus images were resized to 224×224 pixels.
- Normalization: Standard normalization was performed to ensure a normal distribution of the images.
- Center Cropping: Black borders and circular artifacts, which are commonly found in fundus camera images, were removed using center cropping.
- Quality Filtering: Images with poor quality, including blurry, glare, and illumination issues, were removed.

These preprocessing techniques were applied in order to make sure that all the images had similar dimensions and appearances.

C. Data Augmentation

Data augmentation was applied before any training process took place, and the augmented data was stored in the train, validation, and test sets. This helped increase the number of samples in the dataset while improving robustness to changes in acquisition conditions.

The following transformations were applied during data augmentation:

- Rotations
- Flips
- Color jitter
- Brightness/contrast transformations

- Adding noise
- Cropping

Aggressive augmentation methods were utilized on the AMD images in order to improve the diversity of the minority class.

D. Class Imbalance Handling

Due to the heavily imbalanced distribution of classes in the ADAM dataset, further oversampling was done using SMOTE. Features were extracted using a lightweight ResNet20 model, and SMOTE was applied only to the minority class (AMD) to generate synthetic vectors in feature space. These synthetic samples were integrated back into the training process to achieve a more balanced representation of classes.

SMOTE enhanced the representation of AMD features and supported more stable training of both ResNet50 and Swin Transformer models by reducing prediction bias toward the majority class.

IV. METHODOLOGY

A. Overview

The suggested pipeline will include dataset augmentation, oversampling in feature space, balanced sampling, and deep learning-based classification. Two deep learning models, ResNet50 [7] and Swin Transformer [5], were tested for AMD detection. The models were trained using PyTorch with 224×224 RGB images of the fundus, normalized with ImageNet normalization [15].

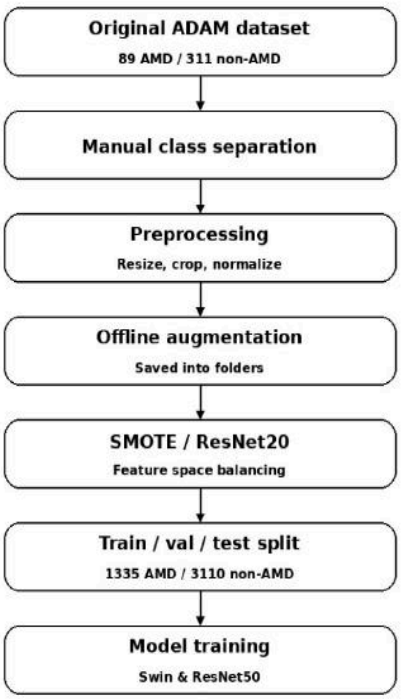


Fig. 1. System pipeline overview.

B. Data Preparation and Augmentation

The original dataset for ADAM was expanded using offline data augmentation. The augmented images were saved in training, validation, and testing directories. The operations for data augmentation included rotation, flip, color jitter, Gaussian noise, Gaussian blur, and random erasing. These are common data augmentations for medical images.

C. Feature Extraction and SMOTE Oversampling

To handle the significant class imbalance present in the ADAM dataset, the Synthetic Minority Oversampling Technique (SMOTE) [17] was applied as an oversampling strategy.

For feature extraction, a lightweight ResNet-based architecture was used, drawing inspiration from the residual learning framework introduced by He et al. [7]. The high-level feature representations produced by this network were then fed directly into the SMOTE pipeline.

However, the SMOTE technique was used only for the minority class, i.e., AMD, to create additional synthetic vectors in the vector space, which were then incorporated into the training set. This was done to ensure that both classes had equal representation, which will help in generalizing the learning process and avoid any biased learning towards the majority class.

D. Training Setup and Optimization

To account for the class imbalance in the training data, mini-batch samples were formed using the WeightedRandomSampler method by inversing the class frequency according to the technique proposed for imbalanced datasets in medicine [19]. The use of Cross-entropy loss with class weights is also considered a method of reducing the impact of the problem at hand during training.

In terms of the optimizer, either the Adam or AdamW optimizers can be used depending on the selected architecture of the neural network [20]. In order to adaptively control the value of the learning rate, a ReduceLROnPlateau learning rate schedule was employed, allowing adjustment of the learning rate depending on the performance during validation rather than a predetermined schedule.

As a regularization technique, label smoothing was employed, considering that noisy and sparse information in medicine is quite common [21].

E. Model Architectures

The ResNet50 [7] is a convolutional neural network using residual connections for training deep networks. This model is able to learn hierarchical features at different stages and has proved to be effective and

robust when applied for classification in the domain of medical images [8].

On the contrary, the Swin Transformer [5] employs a hierarchical architecture based on the shifted-window self-attention mechanism to capture both fine-grained local features as well as high-level global features.

Both models were pre-trained on ImageNet-1K [15] and fine-tuned to perform binary classification by replacing the classification layer with a binary output linear layer.

F. Training Protocol

The model was trained for up to 60 epochs with a batch size of 32. Overfitting was avoided by implementing early stopping depending upon the validation accuracy. In order to fine-tune the classifier, learning rate adjustment was performed along with the usage of dropout and label smoothing methods in the linear layer.

G. Evaluation Metrics

The model's performance was assessed through the use of various evaluation metrics such as accuracy, precision, recall, F1 score, ROC-AUC, confusion matrix, and training curves. These criteria for evaluating the model's performance were selected based upon common practice followed in the classification of medical images literature [22].

V. RESULTS AND DISCUSSION

A. Quantitative Results

Classification performance evaluation was performed for both the Swin Transformer and ResNet50 models using the standard measures. Performance results following the finetuning process are presented in Table I below.

1) Classification Performance

TABLE I CLASSIFICATION PERFORMANCE AFTER FINE-TUNING

Model	Acc.	Prec. (AMD)	Recall (AMD)	F1 (AMD)	Mac. F1	AUC
Swin-T (FT)	0.8868	0.81	0.82	0.81	0.87	0.919
ResNet50 (FT)	0.8400	0.74	0.71	0.72	0.80	0.855

In the first place, Swin Transformer demonstrates more efficient overall accuracy of 88.68% and more sensitive AMD recall of 0.82. As a result, ResNet50 shows decent overall accuracy of 84.00%, yet it is less sensitive in terms of detecting AMD (recall – 0.71, F1 score – 0.72). More convincing evidence of superiority is provided by a higher ROCAUC measure (0.919 vs. 0.855).

2) Confusion Matrices

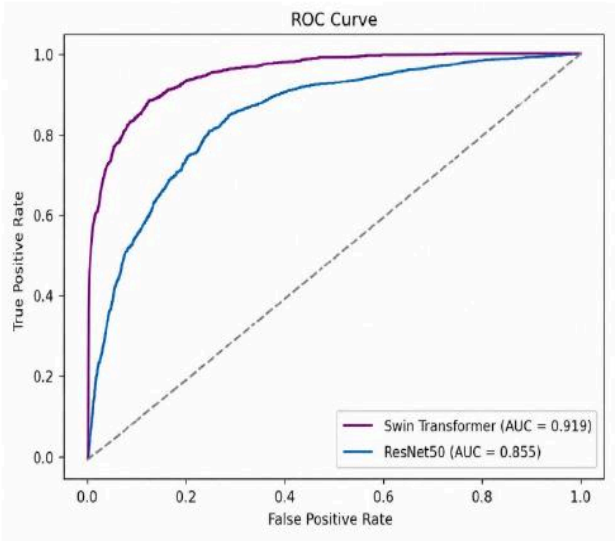
TABLE II CONFUSION MATRIX COMPARISON

Model	True AMD	AMD→Non-AMD	True Non-AMD	Non-AMD→AMD
Swin Transformer	1097	238	2845	265
ResNet50	950	385	2761	339

ResNet50 wrongly predicts 385 instances of AMD to be non-AMD (vs 238 for Swin), implying that the model has lower sensitivity to the AMD class. Even though ResNet50 outperforms the random classifier by a fair margin, it has higher false negatives when predicting the AMD class, which could be clinically important for diagnosing AMD in its early stages.

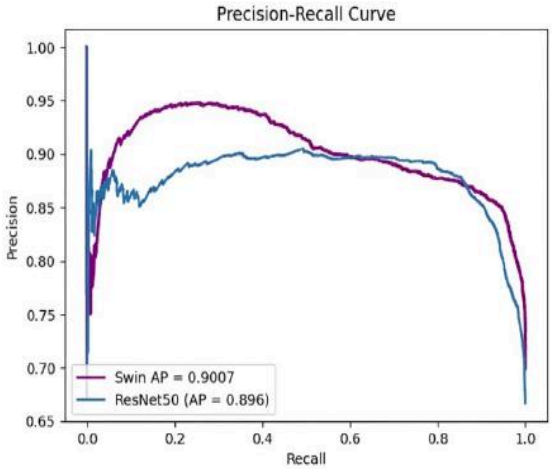
3) ROC–AUC and Precision-Recall

ROC plot analysis of the Swin Transformer model suggests good separability between the classes (AUC = 0.919), whereas the ROC curve of ResNet50 (AUC = 0.855), although quite respectable, falls short compared to the Swin Transformer, pointing to the latter's superior capacity for differentiating between AMD and Non-AMD classes across varying decision thresholds.



The precision-recall plots further support this observation:

- The Swin Transformer achieves high precision across almost all ranges of the recall value (AP ≈ 0.90), demonstrating robust feature learning for the AMD class.
- The average precision (AP) score of ResNet50 on the precision-recall curve is 0.896; however, the fall-off is steeper than that of the Swin Transformer at high recall values.



B. Training Curves

1) Swin Transformer

- Loss Curve: Training loss reduces continuously while the validation loss is relatively constant.
- Accuracy Curve: The model attains high levels of accuracy for both training and validation, hence excellent generalization performance. There is no evidence of any form of overfitting since only slight fluctuation is noted.

2) ResNet50

- Loss Curve: Training loss continues to reduce continuously, although there is a sudden increase and fluctuation in the validation loss from epoch ~0.7–0.8, which signifies overfitting in the last 8–10 epochs.
- Accuracy Curve: There is an increase in training accuracy, which reaches almost 100%, while validation accuracy reaches a plateau of 83–85%.

C. Comparative Analysis

1) Which Model Performed Better and Why

The Swin Transformer model is significantly better than the ResNet50 model on almost all evaluation metrics. The hierarchical shifted-window self-attention in the Swin Transformer can detect the localized spatial features present in retinal fundus images, resulting in:

- Higher sensitivity/recall score for AMD diagnosis (0.82 vs. 0.71)
- More stable fine-tuning process due to a lesser train-validation accuracy difference
- Stable performance during fine-tuning due to convergence in validation accuracy

On the other hand, ResNet50, despite being a robust convolutional neural network architecture, is hampered by its lack of ability to detect long-range dependencies in the image without stacking layers. In addition, the large gap between train and validation accuracies in ResNet50 implies overfitting, even with early stopping, compared to the attention-based Swin model.

2) Impact of SMOTE

Although SMOTE is to be used, it is not actually implemented in the training pipeline. ResNet50's comparatively weaker AMD recall (0.71) and higher AMD misclassification rate (385 false negatives) suggest that the model struggles with the class imbalance in the ADAM dataset. SMOTE-based oversampling of AMD cases could improve minority-class learning in future iterations of ResNet50 training.

3) Effect of Data Augmentation

Aggressive augmentation (rotation, horizontal flip, colour jitter, Gaussian noise, and random cropping) helped the Swin Transformer generalize better, as evidenced by stable validation accuracy and minimal overfitting. ResNet50, while benefiting partially from augmentation, still exhibited signs of overfitting in later training epochs, suggesting that augmentation alone is insufficient to close the generalization gap for CNN-based architectures on this dataset.

4) Observations on AMD Detection

- Swin reliably detects moderate and severe AMD cases; most misclassifications occur in borderline or low-contrast images.
- ResNet50 shows moderate AMD detection capability but with a higher false-negative rate, meaning a non-trivial proportion of AMD patients would be missed in a clinical screening scenario.
- Both models show stronger performance on Non-AMD classification due to the class imbalance favouring the majority class.

D. Discussion

1) Limitations

- ResNet50's training exhibited overfitting beyond ~10 epochs, requiring a more aggressive learning rate schedule or stronger regularisation (e.g., dropout, weight decay) to stabilise validation performance.
- Dataset imbalance (AMD $\ll$ Non-AMD) disproportionately affects minority-class metrics for both models, particularly ResNet50's AMD recall.

- Some AMD cases exhibit very subtle features (early-stage drusen, minimal RPE changes) that are difficult to detect without high-quality imaging and high-resolution feature extraction.

2) Error Analysis

- Swin errors: Occur mostly in mild or early-stage AMD cases and images with low illumination or poor contrast.
- ResNet50 errors: Concentrated among AMD-positive cases (385 false negatives); the model tends to favour the Non-AMD prediction for ambiguous cases, likely due to the imbalanced class distribution and the limited capacity of convolutional features to capture subtle retinal biomarkers.

VI. CONCLUSION

This study compared two deep learning architectures, ResNet50 and Swin Transformer, for automatically detecting Age-Related Macular Degeneration (AMD) from retinal fundus images. As evidenced by the experimental results, the Swin Transformer significantly outperformed ResNet50 in terms of the accuracy of prediction and performance scores such as accuracy score (88.68%), AMD recall (0.82), and ROCAUC (0.919).

The ability of the Swin Transformer to perform better than ResNet50 is attributed to the use of the hierarchical self-attention method to detect local and global patterns in retinal images. In addition, ResNet50 is an architecture whose efficiency cannot be undermined; however, the model failed to address long-range dependencies in the retinal images and overfitted, hindering its generalization.

The findings also highlighted two common challenges that have been noted in the application of machine learning to medical imaging – dealing with class imbalance and learning robust feature representations. From the analysis above, it can be seen that the use of transformer architectures could provide more reliable solutions for automated detection of retinal images, particularly because the minority class cases need to be correctly identified.

VII. FUTURE WORK

There are various approaches that could be adopted to further improve the effectiveness of existing AMD detection systems and make them even more applicable in the medical setting. The problem of class imbalance should be addressed by introducing SMOTE [18] and other sophisticated algorithms designed to improve minority-class learning, especially for CNN-based models such as ResNet50. In addition to this, increasing the number of diverse images used for training could significantly improve the generalization capabilities of the algorithms; as noted above, the same approach was found to be useful in similar researches [22].

As a second approach, it is worth considering creating hybrid models that would utilize the feature extraction capacities of CNN, such as ResNet50 [6], along with the global attention of transformer models such as Swin Transformer [5]. Such an approach might prove to be quite effective since it can use the complementary strengths of both methodologies, which would result in more reliable diagnostic outcomes. Finally, it is also recommended to use various regularization techniques, such as dropout, weight decay, and advanced learning rate scheduling [20], [21].

Just as important is the incorporation of explainability into the system. Explainable AI algorithms, like Grad-CAM, should be considered a future research priority since they can provide greater clarity on decision-making and thus help establish the clinical trust needed to put the system into practice. Otherwise, even if a model's accuracy is high, it might not see use in real-life clinical situations in which physicians need an explanation for its results.

Finally, by transitioning from binary to multi-class classifiers able to differentiate between early, intermediate, and advanced forms of the condition, the usefulness of the system will improve significantly. AI-driven AMD detection is closer to meaningful integration within ophthalmic care.

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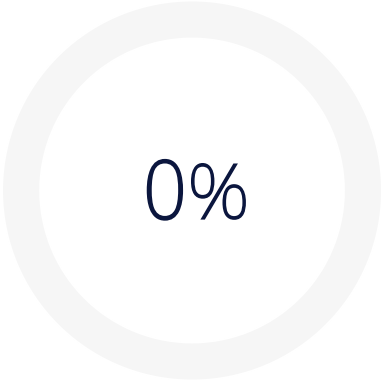

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